SmartSupport AI

KPIs, User Stories & Use Cases

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Scope of this document

LLM provider abstraction, Semantic Kernel orchestration, prompt and intent handling, conversation context, RAG over approved knowledge, tool-calling contracts, safe fallbacks, and their tests. The AI never accesses the database directly; every action is a request to an allowlisted backend tool (BR01, BR02).

Area	Requirements covered (from the FRS v1.0)
Knowledge and RAG	FR-012, FR-013, FR-014 (consumer side), FR-053
Orchestration and context	FR-007, FR-008, FR-010 (Could), FR-016, FR-017
Tool contracts	FR-019 to FR-026
Fallback and handoff support	FR-018, FR-027, FR-028, FR-032
Configuration	FR-048 (provider and prompt settings)

Targets marked plan come from the project plan. All other targets are proposed and must be calibrated with fixtures (assumption A06).

1. KPIs

#	KPI	Target	How measured	FR
K1	Grounded answer accuracy	90% or higher	Policy question test set; answer matches approved article	FR-012
K2	Retrieval hit rate	85% or higher	Correct article in top 3 results	FR-012, 013
K3	Safe handling of unsupported requests	100%	No evidence or out of scope leads to clarify or escalate, never an invented policy	FR-012, 018
K4	Tool selection accuracy	90% or higher	Intent fixtures mapped to the expected tool	FR-016, 019
K5	Invalid tool requests blocked	100%, zero mutations	Unknown tool, forged ID, bypass attempt	FR-019
K6	Response speed (plan)	First token in 2 s; full reply in 8 s	Latency logs under test load	FR-005
K7	Cached answer speed (plan)	500 ms or less	Redis hit latency	FR-053
K8	Fallback availability	100%	AI outage still gives a ticket reference	FR-018

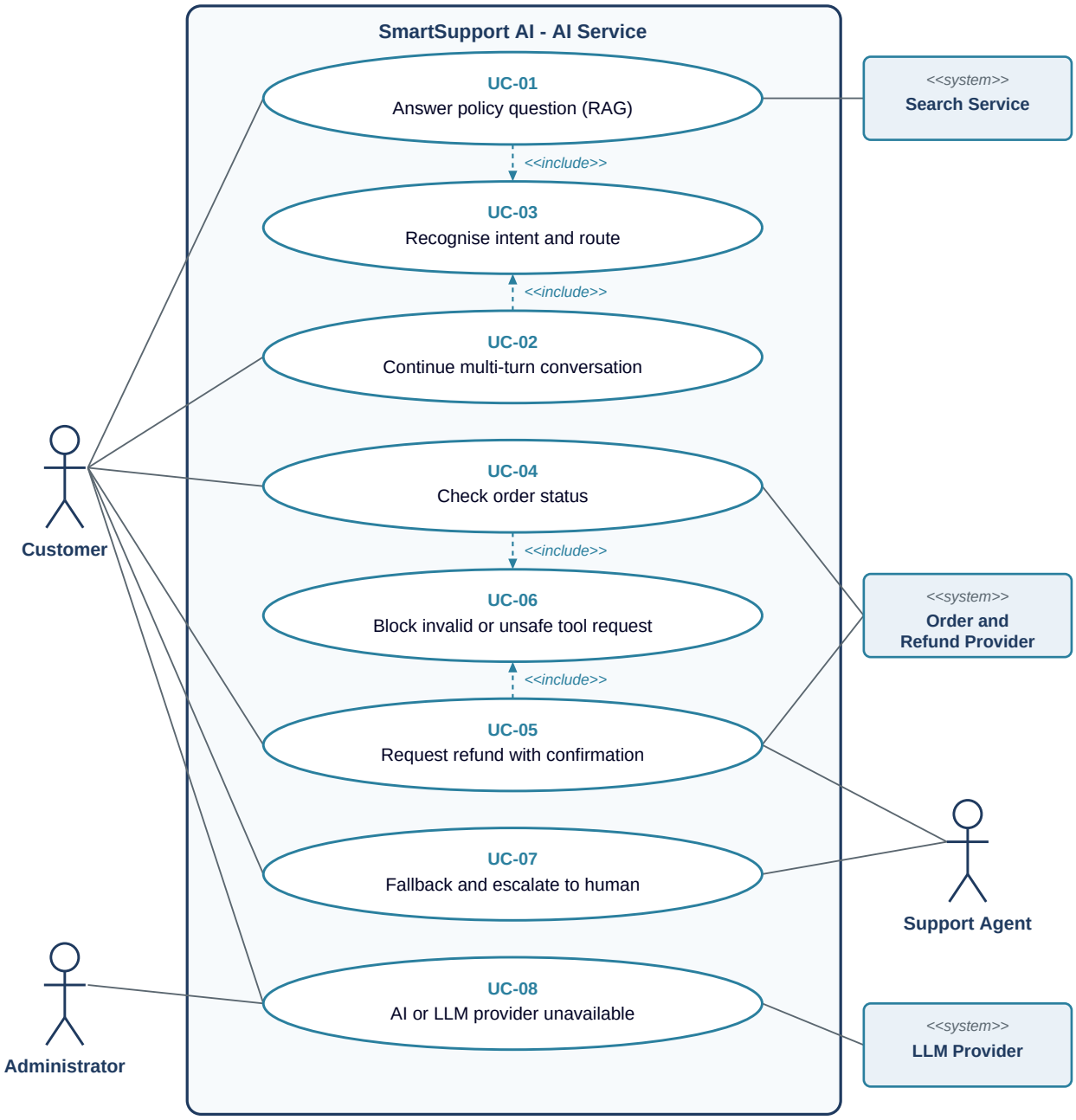
2. User Stories

ID	Story	Acceptance (short)	FR / Sprint
US-01	As a **customer**, I want answers about returns and shipping, so I do not wait for an agent.	Answer matches the approved article and shows its reference.	FR-012 S1
US-02	As a **customer**, I want the assistant to remember earlier turns, so I do not repeat details.	Follow-up uses the same order; no data from other customers.	FR-007 S2
US-03	As a **customer**, I want my order status, so I know when it arrives.	Matches order source; missing fields shown as unavailable.	FR-020 S2
US-04	As a **customer**, I want to request a refund and confirm the exact details first.	Request prepared only after eligibility check and explicit confirmation.	FR-021, 022 S2

ID	Story	Acceptance (short)	FR / Sprint
US-05	As a **customer**, I want to be asked a question when my request is unclear.	Ambiguous input triggers clarification, no unrelated action.	FR-016 S2
US-06	As a **customer**, I want a human when the AI cannot help.	Low confidence or request for a person triggers escalation with a reason.	FR-018, 027 S3
US-07	As a **support agent**, I want an AI summary at handoff, so I understand the case quickly.	Summary, tool results and reason shown; transcript shown if summary fails.	FR-028 S3
US-08	As a **support agent**, I want AI draft replies I can edit.	Drafts stay private until approved.	FR-032 S3
US-09	As an **administrator**, I want to change the prompt or LLM provider without code changes.	New version applies to new requests; rollback works.	FR-048 S4
US-10	As a **tester**, I want repeatable tests for retrieval, unsupported requests and tool selection.	Tests run with mock LLM and fixed data; results recorded.	FR-012, 016, 019

3. Use Cases

3.1 Use Case Diagram



Solid line = actor takes part. Dashed arrow = included use case. UC-03 (intent routing) and UC-06 (tool validation) run inside the AI service for the use cases that include them; the system boxes are external services.

3.2 Use Case Specifications

UC-01 Answer a policy question (RAG)	
Actor	Customer; AI service; Search service
Precondition	Published articles are indexed; chat session is open.
Main flow	1. Customer asks a returns or shipping question. 2. Orchestrator classifies intent as knowledge. 3. SearchKnowledgeBase retrieves published, authorized articles. 4. LLM answers using only retrieved text. 5. Reply streams with article reference; retrieval evidence is logged.
Alternate / exceptions	**3a.** No suitable article or conflicting articles: ask a clarifying question or escalate. **4a.** LLM error: mark reply incomplete and offer retry or escalation.
Postcondition	Customer receives a grounded answer; evidence recorded for review.
Trace	FR-012, FR-013 | BR06

UC-02 Continue a multi-turn conversation	
Actor	Customer; AI service
Precondition	Earlier turns exist in the session.
Main flow	1. Customer sends a follow-up such as "when will it arrive?". 2. System loads the bounded context window with verified tool outcomes. 3. Orchestrator resolves the reference to the previous order. 4. AI replies without asking for repeated details.
Alternate / exceptions	**2a.** Context budget reached: use the conversation summary, keep link to full history. **3a.** Reference missing: ask for the missing detail. **4a.** Summary fails: reduce context safely; never invent a tool result.
Postcondition	Answer is consistent with prior turns; no other customer's context used.
Trace	FR-007, FR-008 | BR01

UC-03 Recognise intent and route	
Actor	AI service
Precondition	Customer message received.
Main flow	1. Classifier returns intent and confidence. 2. Chain of Responsibility passes the request to the matching handler (order, refund, account, knowledge). 3. Strategy selects the response approach for that intent. 4. Decision and confidence are recorded.
Alternate / exceptions	**2a.** Ambiguous intent: clarification or fallback handler; no action executed.
Postcondition	Correct handler is used; decision is traceable.
Trace	FR-016 | BR02, BR06

UC-04 Check order status via tool	
Actor	Customer; AI service; Tool gateway; Order source
Precondition	Customer is authenticated.
Main flow	1. Customer asks about an order. 2. AI requests CheckOrderStatus(orderId). 3. Gateway validates tool, parameters, and ownership. 4. Order source returns status; result is recorded. 5. AI explains status and last update.
Alternate / exceptions	**3a.** Foreign or unknown order: non-disclosing failure message. **4a.** Provider outage: tell the customer status is unavailable; never guess.
Postcondition	Customer sees an accurate status; action record created.
Trace	FR-019, FR-020, FR-026 | BR01, BR02

UC-05 Request a refund with confirmation	
Actor	Customer; AI service; Tool gateway; Approving agent
Precondition	Customer is authenticated and owns the order.
Main flow	1. Customer asks for a refund and gives a reason. 2. AI requests ProcessRefundRequest(orderId, reason). 3. Backend checks eligibility and returns exact amount and currency. 4. AI shows the details and asks for explicit confirmation. 5. On confirmation the backend executes, or sends to human approval if over threshold. 6. AI reports the real status and provider reference.
Alternate / exceptions	**3a.** Ineligible, fully refunded, or foreign order: explain reason, no execution. **5a.** Cancel, changed amount, or expired confirmation: stop and require fresh confirmation. **5b.** Provider timeout: show pending; reconcile before any retry.
Postcondition	Refund recorded once under an operation key; no unconfirmed payment reported as complete.
Trace	FR-021 to FR-023 | BR03, BR04, BR05

UC-06 Block an invalid or unsafe tool request	
Actor	AI service; Tool gateway
Precondition	Model output proposes a tool call.
Main flow	1. Gateway checks the tool is on the allowlist. 2. Parameters are type-validated; identity is taken from the session, not the model. 3. Permissions and business state are checked. 4. Request is rejected with a structured error; denial is logged.
Alternate / exceptions	**1a.** Unknown tool or instruction to bypass policy: reject, no mutation. **2a.** Forged customer ID: reject.
Postcondition	Business state unchanged; denied attempt is auditable.
Trace	FR-019, FR-026 | BR01, BR02

UC-07 Fallback and escalate to a human	
Actor	Customer; AI service; Support agent
Precondition	AI cannot answer reliably, or the customer asks for a person.
Main flow	1. System detects low confidence, repeated tool failure, negative sentiment, or an explicit request. 2. AI offers clarification or escalation. 3. Conversation enters the human queue with a reason. 4. Handoff includes summary, tool results, and unresolved items.
Alternate / exceptions	**1a.** Explicit human request: escalate without any confidence check. **3a.** Repeat escalation: return the existing queue item. **4a.** Summary fails: show transcript and reason; mark summary unavailable.
Postcondition	Agent receives full context; customer is told the true state.
Trace	FR-018, FR-027, FR-028 | BR08

UC-08 AI service or LLM provider unavailable	
Actor	Customer; System; Administrator
Precondition	Provider call fails or times out.
Main flow	1. Provider abstraction reports the failure. 2. System marks the reply incomplete and records a correlated error. 3. Customer is offered retry and a support ticket form. 4. Ticket is created without AI generation and its reference is returned. 5. Administrator may switch provider through versioned configuration.
Alternate / exceptions	**5a.** Invalid configuration: rejected, active version unchanged.
Postcondition	Customer always has a path to help; provider can be replaced without code changes.
Trace	FR-018, FR-048, FR-054 | BR05